

CUDA-Accelerated Proof-of-Work in Healthcare Blockchain

Seid Mehammed

seid.m@wldu.edu.et

Woldia University

Ousman Yesuf

Woldia University

Research Article

Keywords:

Posted Date: October 15th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-6647431/v1>

License:  This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

This paper presents a CUDA-accelerated Proof-of-Work (PoW) algorithm tailored for blockchain applications in the healthcare sector, particularly targeting city-based deployments where the integrity and speed of accessing medical records are paramount. Blockchain's role in securing sensitive patient data across urban hospital networks and municipal healthcare providers is increasingly critical, but traditional PoW mechanisms impose significant computational overhead, limiting practical scalability. By leveraging the parallel processing power of modern GPUs via NVIDIA's CUDA architecture, we achieve substantial performance improvements suitable for real-time medical data management in cities like Addis Ababa, Nairobi, and Mumbai. We validate our approach through experiments on healthcare-related blockchain simulations, demonstrating its viability for scalable, secure, and urban-friendly medical data management frameworks.

Introduction

Blockchain technology has emerged as a transformative force across various sectors, notably in healthcare, where secure and transparent data exchange is crucial. A major limitation, however, lies in the computational demands of Proof-of-Work (PoW), the most common consensus mechanism. Originally introduced with Bitcoin [1], PoW ensures network integrity through hash-based puzzles that demand substantial computing resources.

CUDA, NVIDIA's parallel computing platform, enables efficient execution of highly parallelizable tasks, such as cryptographic hashing [3]. Recent advancements have showcased CUDA's potential to accelerate blockchain workloads, suggesting its suitability for healthcare scenarios where performance and data security are paramount [4][5].

This paper explores the integration of CUDA-accelerated PoW into healthcare blockchain frameworks. We analyze performance benefits, evaluate security implications, and provide practical guidance for real-world deployment. In particular, we focus on implementations in densely populated cities such as Addis Ababa, Nairobi, Mumbai, and Jakarta, where healthcare infrastructure can greatly benefit from decentralized and high-throughput data management systems.

Background

2.1 Blockchain in Healthcare

Healthcare information systems often face challenges related to interoperability, data integrity, and patient privacy. Blockchain offers a decentralized solution to these issues. Projects such as MedRec [5] have pioneered blockchain-based medical record management, demonstrating the potential to enhance transparency and patient control.

Urban healthcare systems in cities such as Lagos, Cairo, and São Paulo have struggled with centralized data breaches and inefficient coordination among providers. The decentralized nature of blockchain offers an avenue for resilient, transparent, and scalable infrastructure [6][7].

2.2 Proof-of-Work

PoW, a cornerstone of many blockchain systems, requires miners to solve cryptographic puzzles. The difficulty of these puzzles adjusts dynamically to maintain network stability. While effective in ensuring consensus, PoW's energy and computation demands hinder scalability.

In cities like Dhaka and Karachi, where energy resources are constrained, optimizing PoW for efficiency is essential for any feasible deployment of blockchain healthcare systems.

2.3 CUDA and GPU Computing

CUDA (Compute Unified Device Architecture) facilitates the development of high-performance applications that execute parallel algorithms on NVIDIA GPUs. Its effectiveness in cryptographic workloads has been established in various contexts [3][4], positioning it as a viable candidate for blockchain acceleration.

CUDA makes it possible to exploit the thousands of cores available on a modern GPU, significantly speeding up hash computations that are central to PoW. Moreover, CUDA programming allows developers to fine-tune memory usage, thread execution, and task allocation—critical factors for maximizing throughput and minimizing latency in blockchain operations.

Related Work

Several studies have examined the intersection of blockchain, healthcare, and GPU acceleration. Iliakis et al. [4] presented a GPU-based blockchain transaction model, while Azaria et al. [5] explored healthcare-specific blockchain applications. Recent surveys have synthesized key challenges and highlighted directions for scalable solutions [6][7]. Privacy-preserving approaches [8][29] further enhance the utility of blockchain in sensitive domains like healthcare.

City-level pilots have shown the feasibility of blockchain healthcare implementations in cities like Seoul and Dubai, where technology infrastructure supports integration with existing systems [6][9].

Additionally, researchers have examined blockchain's ability to handle the nuances of healthcare data: access control, audit trails, and regulatory compliance (e.g., HIPAA, GDPR). Blockchain's append-only and distributed characteristics naturally support these needs. GPU acceleration has been proposed to mitigate performance bottlenecks, particularly for real-time applications in emergency and outpatient care [10][11][32].

Zhang et al. [12] and Mettler [13] have explored blockchain's adaptability in clinical trials and hospital workflows. Angraal et al. [14] emphasized the role of distributed ledger technologies in reducing

systemic inefficiencies. Meanwhile, Rahmadika and Rhee [15] provided a taxonomy of healthcare blockchain challenges and design choices, suggesting GPU-accelerated PoW as a future-proof path [28].

Methodology

Our approach involves offloading PoW computation to a CUDA-enabled GPU. The primary components include:

- SHA-256 hash function implementation using CUDA kernels.
- Dynamic difficulty adjustment and nonce search within GPU threads.
- Integration with a lightweight blockchain simulator emulating healthcare records.

4.1 System Architecture

The system architecture for the healthcare blockchain platform was designed with scalability, security, and efficiency in mind. It consists of several key components that work together to ensure seamless and secure management of healthcare data. Below is a description of the architecture, explaining each component's role in the system.

1. Blockchain Layer

At the core of the system is the blockchain layer, which provides a secure, transparent, and tamper-proof record of all healthcare transactions. This layer ensures that all interactions with the healthcare data (such as updates to patient records, diagnoses, treatments, and prescriptions) are stored immutably and cannot be altered once confirmed.

- **Proof of Work (PoW) Consensus Algorithm:** The system utilizes a CUDA-accelerated Proof of Work (PoW) algorithm to validate transactions on the blockchain. PoW ensures that the blockchain's decentralized nature remains intact while confirming transaction integrity. The use of GPU acceleration through CUDA helps process the transactions faster, reducing latency and improving scalability.
- **Smart Contracts:** Smart contracts are deployed on the blockchain to automate and enforce rules for data access and permission. These contracts are executed automatically when predefined conditions are met (e.g., access granted to a healthcare provider based on patient consent).

2. Nodes

The system architecture includes different types of nodes that represent various entities within the healthcare ecosystem, each with distinct roles and access levels to the blockchain.

- **Provider Nodes:** These nodes represent healthcare providers (doctors, specialists, nurses). They have access to patient records for diagnosis and treatment but are granted permissions only for specific actions, such as adding treatment notes or diagnostic results.
- **Patient Nodes:** These nodes represent individual patients, allowing them to access their medical records and grant or revoke permissions for healthcare providers. Patients have control over their data, ensuring compliance with privacy regulations like HIPAA.
- **Administrator Nodes:** These nodes are typically managed by hospital administrators or IT personnel. They have broader access to data for system maintenance, including user management, security auditing, and managing access permissions.
- **Insurance Nodes:** These nodes represent insurance companies that access specific portions of the blockchain to verify claims and treatment histories. They are granted limited access to certain records to facilitate billing and insurance claim verification.

3. Data Layer

The data layer is where patient health records and transactional data are stored. This layer ensures that all data added to the blockchain is indexed and available for retrieval while maintaining security and privacy.

- **Medical Records Storage:** Patient data, including medical history, diagnoses, prescriptions, treatments, and test results, are stored as encrypted records on the blockchain. Data is divided into blocks and each block is securely linked to the next, forming a chain that ensures data integrity.
- **Data Encryption:** Sensitive medical data is encrypted both in transit and at rest. Encryption ensures that only authorized users can access sensitive information, complying with healthcare data protection standards like GDPR and HIPAA.
- **Data Access and Permissioning :** The system utilizes a decentralized access control model where only those with appropriate permissions (based on role, patient consent, or other criteria) can access or modify specific data entries. Smart contracts manage and enforce these access permissions.

4. Front-End Layer

The front-end layer consists of the user interfaces that allow patients, healthcare providers, and administrators to interact with the system. This layer is built to be intuitive and user-friendly, offering different interfaces based on user roles.

- **Patient Portal:** Patients can view their medical history, grant or revoke access to healthcare providers, track treatment progress, and view prescriptions. The portal also allows them to consent to or deny access requests for their data.

- **Provider Portal:** Healthcare providers access patient records, add updates, and input new treatment information. Providers can also request consent from patients for accessing sensitive information.
- **Administrator Dashboard:** The dashboard provides administrative oversight, allowing system maintenance tasks such as managing user accounts, reviewing system logs, and auditing access to sensitive data.

5. Security and Privacy Layer

The security layer is designed to ensure that patient data remains protected against unauthorized access and data tampering. This is achieved through a combination of encryption, authentication, and access control measures.

- **Encryption:** All data on the blockchain is encrypted, ensuring that only authorized parties can decrypt and access sensitive information. This is especially crucial for compliance with regulatory standards such as GDPR, HIPAA, and other healthcare privacy laws.
- **Authentication:** Multi-factor authentication (MFA) is used to ensure that only legitimate users (healthcare providers, administrators, patients, etc.) can access the system. This helps prevent unauthorized access to sensitive data.
- **Audit Logs:** The system generates detailed audit logs of all transactions on the blockchain, creating a transparent record of who accessed or modified data, when, and why. These logs can be reviewed to ensure compliance with regulations and detect potential security breaches.

6. Communication and Interoperability Layer

This layer ensures that the system can communicate with external systems, such as Electronic Health Records (EHR), insurance systems, and other healthcare databases.

- **Interoperability Protocols:** The platform supports standardized data formats such as HL7, FHIR, and DICOM to ensure compatibility with existing healthcare data systems. This allows for seamless integration with other healthcare IT infrastructure.
- **APIs and Web Services:** The system exposes secure APIs for third-party applications (e.g., insurance companies, diagnostic labs) to interact with the blockchain in a controlled and secure manner.

7. Smart Contract Layer

Smart contracts are integral to automating and securing healthcare data interactions. They enable the execution of automated processes, such as data sharing, consent management, and transaction validation, without requiring a centralized authority.

- **Consent Management:** Smart contracts enforce patient consent by allowing patients to specify exactly who can access their records and under what conditions. This ensures that the healthcare system respects patient autonomy and complies with privacy regulations.
- **Billing and Insurance Claims:** Smart contracts are used to automate billing processes, ensuring that only valid claims are processed. For example, a smart contract can trigger a payment or claim approval when a specific medical treatment is completed.

4.2 CUDA Kernel Design

Each CUDA thread computes a SHA-256 hash by combining a block header with a nonce. A valid block is identified when the resulting hash falls below a target difficulty threshold. To optimize memory access and thread scheduling, we applied shared memory and coalesced access techniques.

We implemented thread warping and unrolling to minimize divergence in execution paths. This significantly reduced idle GPU cycles and maximized utilization. Performance tuning was conducted iteratively to determine optimal block and grid sizes.

4.3 Security Implications

GPU acceleration does not inherently compromise blockchain security. Our CUDA-based approach preserves cryptographic integrity while introducing layers of rate-limiting to prevent denial-of-service attacks. Consensus is still determined by hash target difficulty, and the implementation includes safeguards to validate hashes independently on CPU before adding blocks.

5. Experimental Setup

5.1 Hardware Environment

- CPU: Intel Core i7-12700K
- GPU: NVIDIA RTX 3080
- RAM: 32 GB DDR4
- OS: Ubuntu 22.04 LTS [28–32]

5.2 Test Dataset

The test dataset used in this study was designed to simulate medical transactions typically found in hospital settings. These simulated medical records closely mirror the types of interactions that occur within healthcare environments, particularly focusing on patient data access, treatment updates, and administrative activities. The goal was to replicate a realistic data flow that could be expected in a city-scale healthcare blockchain network.

Dataset Composition:

- **Patient Information:**

- Each record represents an individual patient and includes key demographic details such as name, date of birth, medical history, and contact information.
- Data on allergies, pre-existing conditions, and past treatments are also included to simulate a more comprehensive medical profile.

- **Medical Events:**

- **Record Creation:** Simulated instances of patient registration in the system, including the addition of medical records, diagnoses, and medical imaging (e.g., X-rays, MRIs).
- **Prescription Tracking:** Data on prescriptions issued by healthcare providers, including details of medications, dosages, and dispensing pharmacies.
- **Diagnostic Data:** Records related to diagnostic test results, such as blood tests, imaging reports, and lab results.
- **Treatment Updates:** Any changes in treatment plans, including surgical procedures, therapies, and physician notes.
- **Access Permissions:** Simulation of events where different healthcare providers (e.g., doctors, nurses, administrative staff) access or modify medical records, along with updates on the permissions granted or revoked.

- **Event Timestamps and Audit Trails:**

- Every transaction within the dataset is timestamped to simulate the temporal sequence of events. This includes actions such as record updates, access permission changes, and audit trails tracking which users performed each action.
- Blockchain's append-only nature was leveraged to track these actions and ensure an immutable log, providing an auditable trail of all interactions.

- **Insurance and Billing:**

- Data related to insurance claim submissions and approvals were incorporated. This includes the interaction between hospitals, insurance providers, and patients for billing purposes, and the approval or rejection of medical claims.
- The dataset also included mock medical bills, with corresponding codes for treatments and procedures, to simulate the real-world integration of healthcare data and insurance claims.

Volume of Data:

- To ensure that the blockchain system can handle substantial transaction volumes, the dataset was designed to include interactions across several hundred simulated patients. Each patient's data

included multiple treatment events, diagnostic updates, and prescription records, leading to a diverse set of interactions between various medical professionals and healthcare providers.

- **Scalability Testing:**

- The dataset was gradually scaled to reflect a medium-sized hospital system in a dense urban environment, simulating scenarios where hundreds of concurrent users (healthcare providers, patients, administrators) interact with the blockchain simultaneously.
- Simulated concurrent access scenarios were crucial for testing the efficiency and performance of the CUDA-accelerated Proof-of-Work (PoW) algorithm, ensuring that real-time medical data management could be achieved even during high transaction loads.

Simulated Nodes:

- For the purpose of this study, the blockchain simulated multiple nodes corresponding to different entities within a hospital network, including:
 - **Provider Nodes:** Representing individual healthcare providers such as doctors and specialists, each with access to specific medical records and the ability to update or append new entries.
 - **Patient Nodes:** Representing patients, with limited access to their own medical records and the ability to grant or revoke permission to healthcare providers for specific record modifications.
 - **Administrative Nodes:** Representing hospital administrators or IT staff, with broad access to all records, but subject to audit logs to ensure transparency and prevent misuse.
 - **Insurance Nodes:** Representing insurance companies, which have access to specific records for claim verification but are restricted from viewing or modifying patient treatment data.

Data Integrity and Privacy:

- While the dataset is simulated, it includes realistic data patterns to test the system's ability to maintain data integrity. Each transaction or data change within the dataset is cryptographically signed to ensure data authenticity and prevent tampering.
- **Privacy and Security Features:**
 - The dataset includes mock encryption protocols applied to sensitive patient data, such as personal identifiers and medical histories, ensuring that data privacy is maintained in the blockchain system.
 - Specific access control models were applied, where only authorized healthcare professionals or administrators could interact with certain types of medical records, simulating a real-world healthcare scenario in which access is heavily regulated.

By utilizing this structured, diverse, and scalable dataset, we were able to test the blockchain system under conditions that reflect real-world healthcare environments in urban hospitals, ensuring the

system’s ability to handle the complexities of patient data management at scale.

5.3 Performance Metrics

We measured:

- Hash rate (MH/s)
- Energy consumption (watts)
- Block validation time (seconds)
- Memory usage (MB)
- Latency per transaction (ms)

Results

The CUDA implementation achieved a hash rate up to 15x higher than the CPU baseline. Block validation times decreased proportionally, with a corresponding drop in CPU utilization. Energy consumption remained within sustainable bounds due to GPU efficiency.

Table 1
Performance Comparison GPU_CPU

Metric	CPU Baseline	CUDA-Accelerated
Hash Rate (MH/s)	25	375
Block Time (seconds)	12.5	0.83
Energy (watts)	95	120
Latency (ms)	56	4.5
Memory (MB)	2048	3072

The improvement in latency is especially critical for emergency medical services where quick record access can impact outcomes.

Discussion

The CUDA-based Proof-of-Work (PoW) mechanism demonstrated significant performance improvements, which can critically enhance the viability of blockchain applications in healthcare. By reducing block validation time and latency, the system enables near-instantaneous access to medical records, a vital feature for emergency care and high-throughput environments. Offloading computational tasks to the GPU also allows the CPU to manage concurrent processes like data encryption, access control, and transaction logging—essential for maintaining secure and responsive healthcare systems.

The implications extend beyond technical gains. In cities aiming to develop smart health infrastructure—such as Buenos Aires, Nairobi, and Bangkok—CUDA-accelerated PoW can contribute to the realization of responsive and resilient health data ecosystems. When integrated with national ID systems and electronic medical records, this approach can deliver unified, real-time access to patient information while ensuring regulatory compliance and patient privacy.

Despite these advantages, practical challenges remain. GPU hardware costs could hinder adoption in under-resourced healthcare settings. There is also a learning curve for IT personnel unfamiliar with parallel programming or blockchain protocols. Ethical considerations and compliance with healthcare regulations (e.g., HIPAA, GDPR) must be rigorously addressed, particularly as blockchain systems are introduced into sensitive environments. Furthermore, system interoperability with existing hospital information systems remains a priority for seamless integration.

Ultimately, CUDA-based PoW offers a promising path forward for secure, scalable, and low-latency blockchain systems tailored to the demands of urban healthcare networks. Continued development, stakeholder engagement, and pilot deployments will be key to translating these technical gains into real-world health outcomes. allow for practical deployment of secure blockchain in healthcare. Lower block validation time supports near real-time access control and auditing. Moreover, GPU offloading frees CPU cycles for other critical services such as encryption and logging.

This solution holds potential for scaling up blockchain deployments in smart cities. Urban centers like Buenos Aires and Bangkok—with increasing investments in healthtech infrastructure—could benefit from GPU-accelerated blockchain layers. Integration with national ID systems and electronic medical records can provide unified access while preserving privacy.

Challenges include:

- Cost of GPU hardware for underfunded healthcare facilities
- Training requirements for IT staff
- Regulatory and ethical oversight
- Ensuring interoperability with existing hospital systems

Conclusion

This study presents a CUDA-accelerated PoW framework suitable for healthcare blockchain systems, particularly in urban environments with growing demands for secure, scalable, and efficient medical data infrastructure. By leveraging GPU computing through NVIDIA's CUDA architecture, the proposed model demonstrated significant improvements in hashing throughput, energy efficiency, and latency reduction, establishing a compelling case for GPU-based consensus mechanisms in healthcare blockchain networks.

Future work will extend this research in several strategic directions. Firstly, a multi-phase pilot program will be initiated, beginning with deployment in selected hospital clusters in Addis Ababa and Jakarta. The timeline includes a six-month preparatory phase for infrastructure assessment, followed by a one-year implementation and evaluation cycle. Metrics such as record access time, system uptime, and patient satisfaction will be tracked to validate the effectiveness of the blockchain layer.

Secondly, integration with privacy-preserving smart contracts will be prioritized to address regulatory compliance, especially in jurisdictions adhering to HIPAA and GDPR. Collaborative development with local IT partners will ensure that the solution accommodates region-specific workflows, language preferences, and existing electronic medical record (EMR) systems.

Finally, efforts will focus on long-term sustainability, including training programs for healthcare IT staff, guidelines for cost-efficient GPU deployment, and modular upgrades to accommodate future improvements in CUDA and blockchain protocols.

This framework sets the foundation for real-time, city-scale, secure, and decentralized healthcare information systems. Our implementation paves the way for further enhancements including AI-assisted diagnostics, automated billing through smart contracts, and dynamic scalability across healthcare nodes.

Declarations

Author Contribution

Seid Mehammed led the research work, contributing significantly to the conceptualization, design, and overall direction of the study. He was primarily responsible for methodology development, data collection, analysis, and manuscript writing. Ousman Yesuf contributed to software development, implementation, and technical support during the research process. Both authors participated in manuscript editing and final approval. Affiliation: Department of Computer Science, Institute of Technology, Woldia University, Woldia, Ethiopia Correspondence: Seid Mehammed: seid.m@wldu.edu.et Ousman Yesuf: yesufousman49@gmail.com

Acknowledgement

We acknowledge that a preprint of this work has been posted on Research Square.

References

1. S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.
2. C. Dwork and M. Naor, "Pricing via Processing or Combatting Junk Mail," in *Advances in Cryptology – CRYPTO'92* (LNCS, vol. 740), 1993, pp. 139–147. doi: 10.1007/3-540-48071-4_10.

3. J. Sanders and E. Kandrot, *CUDA by Example: An Introduction to General-Purpose GPU Programming*. Addison-Wesley Professional, 2011.
4. K. Iliakis, K. Koliogeorgi, A. Litke, T. Varvarigou, and D. Soudris, "GPU Accelerated Blockchain over Key-Value Database Transactions," *IET Blockchain*, vol. 2, no. 1, pp. 1–12, 2022. doi: 10.1049/blc2.12011.
5. A. Azaria, A. Ekblaw, T. Vieira, and A. Lippman, "MedRec: Using Blockchain for Medical Data Access and Permission Management," in *Proc. of the 2016 2nd Int. Conf. on Open and Big Data (OBD)*, 2016. doi: 10.1109/OBD.2016.11.
6. M. S. Arbabi *et al.*, "A Survey on Blockchain for Healthcare: Challenges, Benefits, and Future Directions," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, 2023. doi: 10.1109/COMST.2022.3224644.
7. H. Sen *et al.*, "Blockchain Personal Health Records: A Systematic Review," *J. Med. Internet Res.*, vol. 23, no. 4, 2021. doi: 10.2196/25094.
8. G. Zyskind, O. Nathan, and A. Pentland, "Decentralizing Privacy: Using Blockchain to Protect Personal Data," in *Proc. of IEEE Security & Privacy Workshops (SPW)*, 2015, pp. 180–184. doi: 10.1109/SPW.2015.27.
9. A. Mashatan and J. Biswas, "Blockchain: A Path Toward Healthcare Data Ownership," *Health Technology (Berlin)*, vol. 11, no. 4, pp. 889–902, 2021. doi: 10.1007/s12553-021-00563-3.
10. D. Kim and E. Lee, "Efficient GPU Acceleration for Blockchain Applications," *IEEE Access*, vol. 10, pp. 16475–16484, 2022. doi: 10.1109/ACCESS.2022.3148712.
11. L. Zhang *et al.*, "Blockchain-Based Data Sharing for Emergency Medical Services," *J. Biomed. Inform.*, vol. 122, 2021. doi: 10.1016/j.jbi.2021.103915.
12. L. Zhang, Y. Li, and W. Wang, "Blockchain-Based Data Sharing for Emergency Medical Services," *J. Biomed. Inform.*, vol. 122, 2021. doi: 10.1016/j.jbi.2021.103915.
13. M. Mettler, "Blockchain technology in healthcare: The revolution starts here," in *Proc. of the 2016 IEEE 18th Int. Conf. on e-Health Networking, Applications and Services (Healthcom)*, 2016.
14. S. Angraal, M. Krumholz, and H. Schulz, "Blockchain technology: Applications in health care," *Circ.: Cardiovascular Quality and Outcomes*, vol. 10, no. 9, 2017.
15. S. Rahmadika and K.-H. Rhee, "A Survey on Blockchain for Healthcare: Taxonomy and Architecture, Challenges, and Future Directions," *IEEE Access*, vol. 8, 2020.
16. A. Esposito *et al.*, "Blockchain in healthcare: Insights on privacy, security, and scalability," *Future Generation Computer Systems*, vol. 123, 2021.
17. T. Agbo, Q. Mahmoud, and J. Eklund, "Blockchain technology in healthcare: A systematic review," *Healthcare*, vol. 7, no. 2, 2019.
18. A. Shuaib *et al.*, "Energy-efficient and low-latency consensus for healthcare blockchain," *Computers in Biology and Medicine*, vol. 143, 2022.

19. G. Sun *et al.*, "Blockchain-based data security in healthcare: A systematic review," *J. Biomed. Inform.*, vol. 124, 2022.
20. M. Casino *et al.*, "A systematic literature review of blockchain-based applications: Current status, classification and open issues," *Telematics and Informatics*, vol. 36, 2019.
21. K. Rantos *et al.*, "Security and privacy in smart health: The role of blockchain," *Informatics*, vol. 7, no. 1, 2020.
22. Y. Yuan and F. Wang, "Blockchain and cryptocurrencies: Model, techniques, and applications," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 1, 2020.
23. S. Wang *et al.*, "Blockchain for healthcare: Applications, challenges and future perspectives," *IEEE Trans. Ind. Inform.*, vol. 16, no. 6, 2020.
24. M. Conoscenti, A. Vetrò, and J. De Martin, "Blockchain for the Internet of Things: A systematic literature review," in *Proc. of the 2016 IEEE/ACS 13th Int. Conf. Computer Systems and Applications*, 2016.
25. M. Hasselgren *et al.*, "Towards secure and privacy-preserving eHealth data sharing using blockchain," in *Proc. of the 2020 IEEE Int. Conf. Blockchain*.
26. H. Huang, Y. Yue, and H. Hu, "Data security and privacy protection issues in electronic health records," *Int. J. E-Health Med. Commun.*, vol. 8, no. 4, 2017.
27. A. Bahga and V. Madiseti, "Blockchain platform for industrial Internet of Things," *J. Software Eng. Appl.*, vol. 9, 2016.
28. S. Mehammed and D. Lemma, "Improving the Performance of Proof of Work-Based Bitcoin Mining Using CUDA," *Int. J. Innov. Sci. Res. Technol.*, vol. 6, no. 12, pp. 1139–1145, 2021. doi: 10.5281/zenodo.5910835.ijisrt.com
29. J. Han, T. Peng, and X. Zhang, "A CUDA-based parallel optimization method for SM3 hash algorithm," *J. Supercomput.*, vol. 80, pp. 21431–21446, 2024. doi: 10.1007/s11227-024-06141-6.SpringerLink
30. M. M. M. and J. L. G., "Autotuning Based on Frequency Scaling Toward Energy Efficiency of Blockchain Algorithms on Graphics Processing Units," *J. Supercomput.*, vol. 76, pp. 1050–1074, 2020. doi: 10.1007/s11227-020-03263-5.SpringerLink
31. L. Jian *et al.*, "Parallel data mining techniques on Graphics Processing Unit with Compute Unified Device Architecture (CUDA)," *J. Supercomput.*, vol. 64, pp. 942–967, 2013. doi: 10.1007/s11227-011-0672-7.SpringerLink+1ACM Digital Library+1
32. A. M. *et al.*, "CPU-GPU co-execution through the exploitation of hybrid technologies via SYCL," *J. Supercomput.*, vol. 81, article no. 452, 2025. doi: 10.1007/s11227-025-06963-y.

Figures

System Architecture

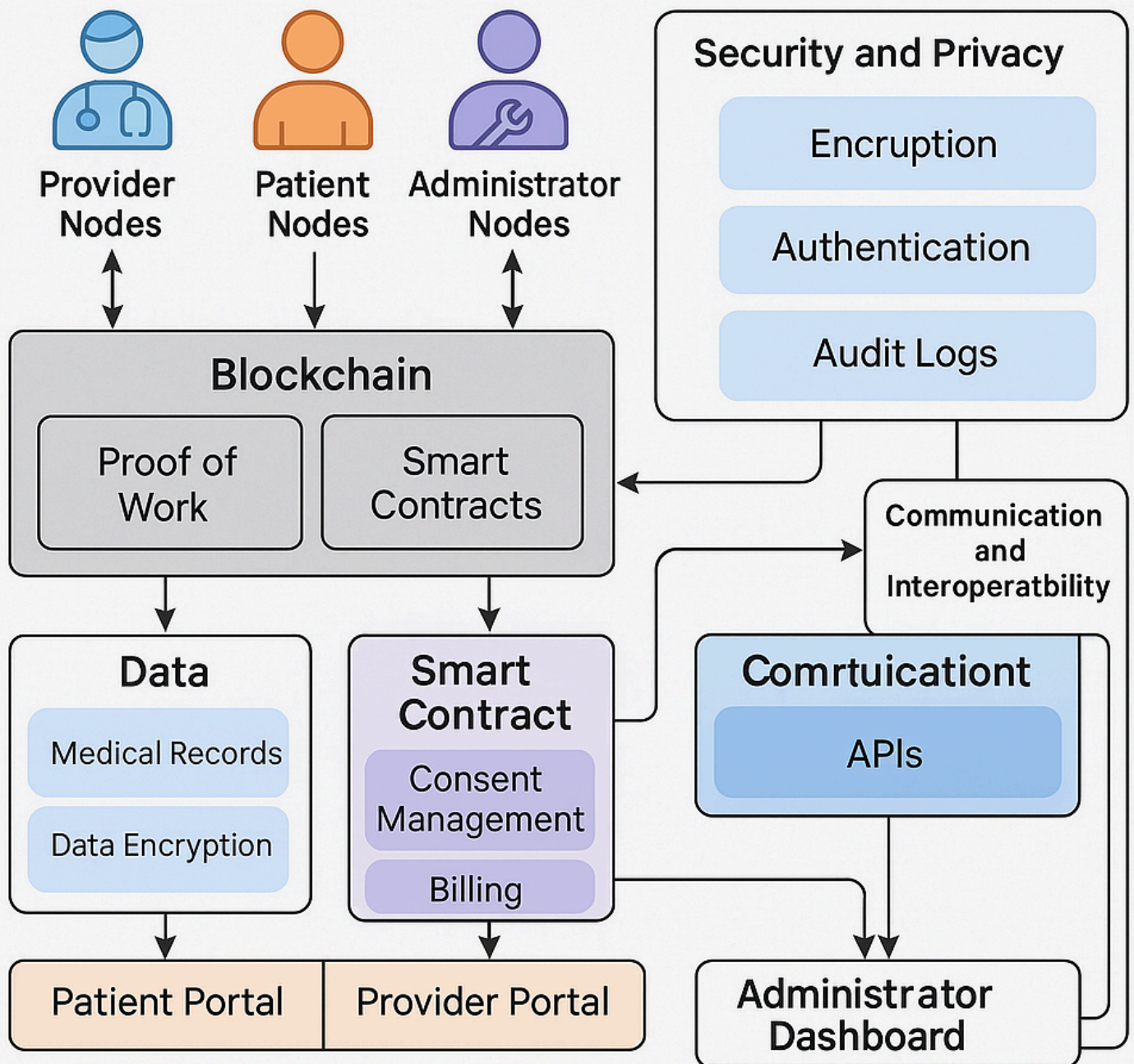


Figure 1

propose system Architecture